

HDOC – Experiment 22

Question

Menu-Driven Program Using Switch-Case

❖ Understanding problem

Input:

- m — positive integer
- n — positive integer, where $m \leq n$
- choice — menu choice

Menu operations:

1. Print multiplication tables of m and n side by side, from 1 to 10.
2. Print a multiplication grid having rows from 1 to m and columns from 1 to n.
3. For every integer from m to n, display that integer and the next five consecutive integers.
4. Exit.

Validation:

- m and n must both be positive.
- $m \leq n$.
- Menu choice must be from 1 to 4.
- Invalid menu choices should display an error, and the menu should appear again.

❖ Algorithm

1. Start.
2. Declare integer variables m, n, choice, i, and j.
3. Input m and n.
4. Check whether $m > 0$, $n > 0$, and $m \leq n$.
5. If the condition is false:
 - Display "Invalid input. Enter positive integers with $m \leq n$."
 - Go back to Step 3.
6. Display the menu:-
 1. Multiplication tables of m and n
 2. Multiplication grid
 3. Consecutive integers
 4. Exit
7. Input choice.
8. Use a switch statement on choice.
9. If choice = 1:
 - Display the heading.
 - Set $i = 1$.
 - Repeat while $i \leq 10$:
 - Print $m \times i = m*i$.
 - Print $n \times i = n*i$ on the same line.
 - Increment i.
10. If choice = 2:
 - Display the multiplication grid heading.

- Print column headings from 1 to n.

- For i = 1 to m:

- Print i as the row heading.
- For j = 1 to n:
 - Print i*j.

11. If choice = 3:

- Display the consecutive integers heading.

- For i = m to n:

- Print i followed by :.
- For j = 0 to 5:
 - Print i+j.

12. If choice = 4:

- Display "Exiting program...".

- Exit the menu loop.

13. For any other choice:

- Display "Invalid menu choice. Please enter a choice from 1 to 4."

14. If choice is not 4, go back to Step 6.

15. Stop.

❖ Test case table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	m=3, n=5, choice=1	Tables of 3 and 5 side by side from $\times 1$ to $\times 10$	Tables of 3 and 5 displayed correctly	Pass
2	m=3, n=5, choice=2	3×5 multiplication grid	Correct multiplication grid displayed	Pass
3	m=3, n=5, choice=3	3: 3 4 5 6 7 8, 4: 4 5 6 7 8 9, 5: 5 6 7 8 9 10	Consecutive integers displayed correctly	Pass
4	m=3, n=5, choice=4	Program exits	Program exited successfully	Pass
5	m=5, n=3	Error because $m > n$	Invalid input message displayed	Pass
6	m=-2, n=5	Error because m is not positive	Invalid input message displayed	Pass
7	m=0, n=5	Error because m is not positive	Invalid input message displayed	Pass
8	m=5, n=5, choice=1	Table of 5 displayed in both columns	Correct output displayed	Pass
9	m=1, n=1, choice=2	Grid containing $1 \times 1 = 1$	Correct grid displayed	Pass
10	m=3, n=5, choice=7	Invalid menu choice message; menu displayed again	Error message displayed correctly	Pass

❖ Program code in C

```
#include <stdio.h>

int main()
{
    int m, n, choice;
    int i, j;

    /* Input and validation of m and n */
    do
    {
        printf("Enter two positive integers m and n (m <= n): ");
        scanf("%d %d", &m, &n);

        if (m <= 0 || n <= 0 || m > n)
        {
            printf("Invalid input. Enter positive integers with m <= n.\n");
        }

    } while (m <= 0 || n <= 0 || m > n);

    /* Menu loop */
    do
    {
        printf("\n1. Multiplication tables of m and n\n");
        printf("2. Multiplication grid\n");
        printf("3. Consecutive integers\n");
```

```
printf("4. Exit\n");
```

```
printf("Enter your choice: ");
```

```
scanf("%d", &choice);
```

```
switch (choice)
```

```
{
```

```
case 1:
```

```
printf("\nMultiplication Tables of %d and %d (m=%d, n=%d)\n\n",
```

```
    m, n, m, n);
```

```
for (i = 1; i <= 10; i++)
```

```
{
```

```
    printf("%d x %d = %-5d\t%d x %d = %d\n",
```

```
        m, i, m * i,
```

```
        n, i, n * i);
```

```
}
```

```
break;
```

```
case 2:
```

```
printf("\nMultiplication Grid (m=%d, n=%d)\n\n", m, n);
```

```
/* Print column headings */
```

```
printf("    ");
```

```
for (j = 1; j <= n; j++)
```

```
{
```

```
    printf("%4d", j);
```

```
}
```

```

printf("\n ");
for (j = 1; j <= n; j++)
{
    printf("----");
}
printf("\n");

/* Print multiplication grid */
for (i = 1; i <= m; i++)
{
    printf("%3d | ", i);

    for (j = 1; j <= n; j++)
    {
        printf("%4d", i * j);
    }

    printf("\n");
}

break;

```

case 3:

```

printf("\nConsecutive Integers from %d to %d (m=%d, n=%d)\n\n",
    m, n, m, n);

for (i = m; i <= n; i++)
{

```

```
printf("%d: ", i);
```

```
for (j = 0; j <= 5; j++)
```

```
{
```

```
    printf("%d ", i + j);
```

```
}
```

```
printf("\n");
```

```
}
```

```
break;
```

```
case 4:
```

```
    printf("\nExiting program...\n");
```

```
    break;
```

```
default:
```

```
    printf("\nInvalid menu choice. Please enter a choice from 1 to 4.\n");
```

```
}
```

```
} while (choice != 4);
```

```
return 0;
```

```
}
```

❖ Compile, Execute and Evaluate

```
(kali㉿kali)-[~/Desktop/exp22]
$ vi exp22.c

(kali㉿kali)-[~/Desktop/exp22]
$ gcc exp22.c -o exp22

(kali㉿kali)-[~/Desktop/exp22]
$ ./exp22
Enter two positive integers m and n 3 5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 1

Multiplication Tables of 3 and 5 (m=3, n=5)

3 x 1 = 3      5 x 1 = 5
3 x 2 = 6      5 x 2 = 10
3 x 3 = 9      5 x 3 = 15
3 x 4 = 12     5 x 4 = 20
3 x 5 = 15     5 x 5 = 25
3 x 6 = 18     5 x 6 = 30
3 x 7 = 21     5 x 7 = 35
3 x 8 = 24     5 x 8 = 40
3 x 9 = 27     5 x 9 = 45
3 x 10 = 30    5 x 10 = 50

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 2
```

```
Multiplication Grid (m=3, n=5)

      1  2  3  4  5
    --
1 |  1  2  3  4  5
2 |  2  4  6  8 10
3 |  3  6  9 12 15

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 3

Consecutive Integers from 3 to 5 (m=3, n=5)

3: 3 4 5 6 7 8
4: 4 5 6 7 8 9
5: 5 6 7 8 9 10

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 4

Exiting program...

(kali㉿kali)-[~/Desktop/exp22]
$ ./exp22
Enter two positive integers m and n (m ≤ n): 5 3
Invalid input. Enter positive integers with m ≤ n.
Enter two positive integers m and n (m ≤ n): -2 5
Invalid input. Enter positive integers with m ≤ n.
Enter two positive integers m and n (m ≤ n): 5 5
```

```

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 1

Multiplication Tables of 5 and 5 (m=5, n=5)

5 x 1 = 5      5 x 1 = 5
5 x 2 = 10     5 x 2 = 10
5 x 3 = 15     5 x 3 = 15
5 x 4 = 20     5 x 4 = 20
5 x 5 = 25     5 x 5 = 25
5 x 6 = 30     5 x 6 = 30
5 x 7 = 35     5 x 7 = 35
5 x 8 = 40     5 x 8 = 40
5 x 9 = 45     5 x 9 = 45
5 x 10 = 50    5 x 10 = 50

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 4

Exiting program...

(kali@kali)-[~/Desktop/exp22]
$ ./exp22
Enter two positive integers m and n (m ≤ n): 1 1

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 2

```

```

Multiplication Grid (m=1, n=1)

  1
  ──
1 | 1

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 4

Exiting program...

(kali@kali)-[~/Desktop/exp22]
$ ./exp22
Enter two positive integers m and n (m ≤ n): 3 5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 7

Invalid menu choice. Please enter a choice from 1 to 4.

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 4

Exiting program...

```